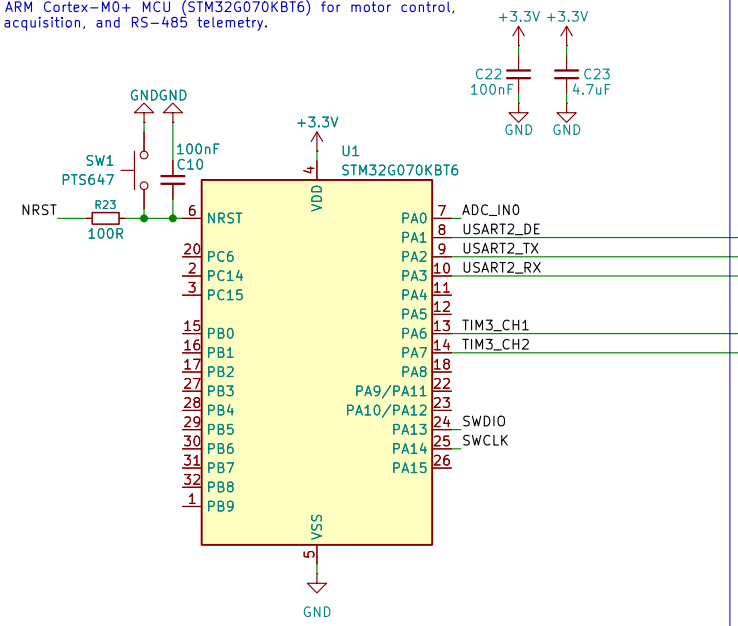
Microcontroller Core & Supervision

32-bit ARM Cortex-M0+ MCU (STM32G070KBT6) for motor control, sensor acquisition, and RS-485 telemetry.



- Clock: Internal RC up to 64 MHz | Supply: +3.3V
- ADC (PA0): Motor current sensing (IPROPI, 0V to 3.3V)
- Reset: Active-low with hardware RC filter (100R / 100nF, τ = 10 µs)

InterfaceSheet

RS485_DE_MCU
RS485_TX_MCU
RS485_RX_MCU
TIM_ENC_A
TIM_ENC_B
TRIG1_MCU
TRIG2_MCU
OPTO1_MCU
OPTO2_MCU
OPTO1_EXT_P
OPTO1_EXT_N
OPTO2_EXT_P
OPTO2_EXT_N
TRIG2_EXT
TRIG1_EXT
RS485_A_EXT
RS485_B_EXT
ENC_B_IN
ENC_A_IN

Fichier: Interface.kicad_sch

DriverMotorSheet

NSLEEP_MCU
NFAULT_MCU
PMODE_MCU
IPROPI_MCU
PH/IN2_MCU
EN/IN1_MCU

Fichier: DriverMotor.kicad_sch

ConnectorSheet

Fichier: Connector.kicad_sch

PowerSheet

Fichier: Power.kicad_sch

BerryRocket

Sheet: /
File: KinetiCoreDC.kicad_sch

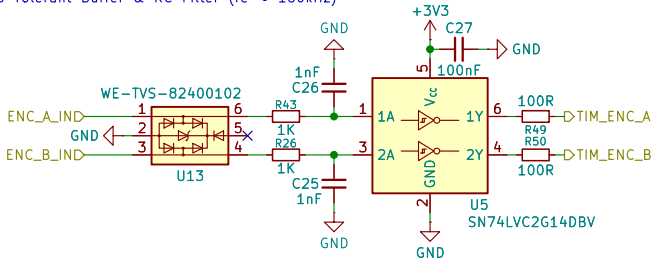
Title: KinetiCoreDC – Main

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	Rev: V°1
KiCad E.D.A. 10.0.2			Id: 1/5

Text

Encoder interface & ESD protection

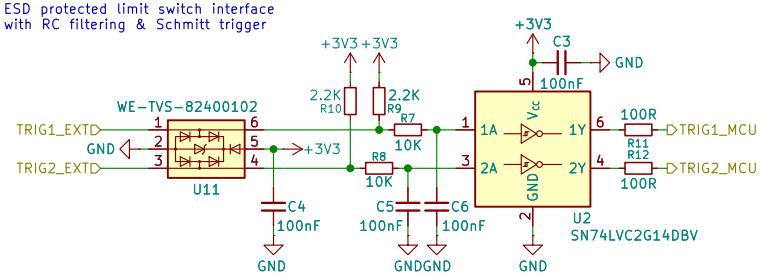
Incremental Quadrature (A/B),
5V/3V3 Tolerant Buffer & RC Filter (fc ≈ 160kHz)



- Input: Quadrature A/B, 3.3V / 5V tolerant
- HF Noise Filter: Low-pass RC (1kΩ / 1nF, fc ≈ 160 kHz, τ = 1 µs)
- Output: Inverted Schmitt-trigger to STM32 TIM3_CH1/CH2

Limit Switch Input Conditioning

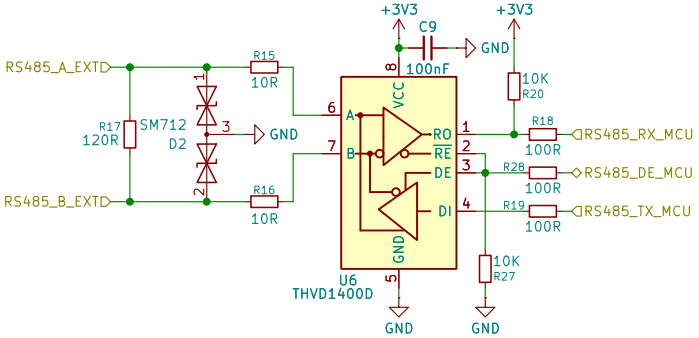
ESD protected limit switch interface
with RC filtering & Schmitt trigger



- Input type: Dry contact to GND (Internal 2.2kΩ pull-up to +3.3V)
- Debounce filter: Low-pass RC (10kΩ / 100nF, fc ≈ 160 Hz, τ = 1 ms)
- Output: Inverted Schmitt-trigger logic (Active-HIGH at MCU when switch closed)

Protected Half-Duplex RS-485

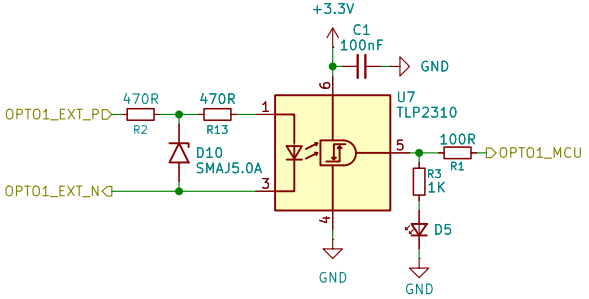
Protected 3.3V half-duplex transceiver stage (up to 500 kbps)



- Bus specs: Half-duplex up to 500 kbps (Slew-rate limited, 120Ω terminated)
- Common-mode protection: −7V to +12V (SM712 TVS + 10Ω series)
- Control: DE/RE tied together (Default RX mode via 10kΩ pull-down)

Optoisolated PWM Input N°1

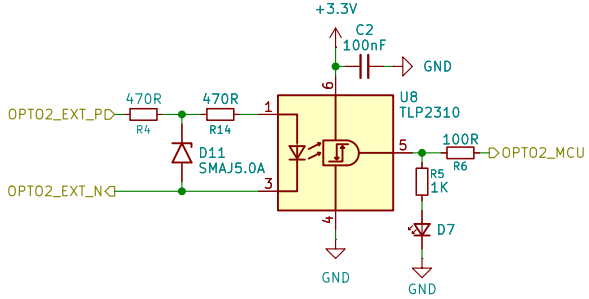
Optoisolated input for secure PWM/GPIO control,
protecting the logic circuit from DC motor noise.



- Input range: 5V to 24V differential (R_series = 940Ω, ~3.5mA @ 5V)
- Max frequency: 5 Mbps (Push-pull output, active-high)
- Protection: Reverse/Overvoltage clamp at 5.0V (SMAJ5.0A)

Optoisolated PWM Input N°2

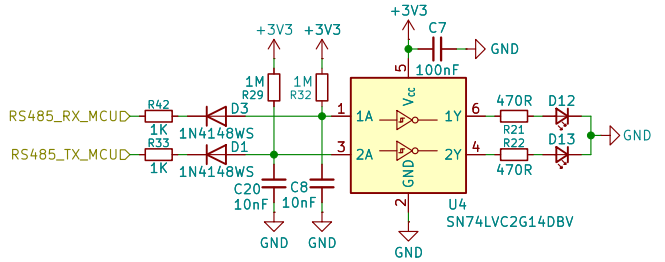
Optoisolated input for secure PWM/GPIO control,
protecting the logic circuit from DC motor noise.



- Input range: 5V to 24V differential (R_series = 940Ω, ~3.5mA @ 5V)
- Max frequency: 5 Mbps (Push-pull output, active-high)
- Protection: Reverse/Overvoltage clamp at 5.0V (SMAJ5.0A)

UART activity LED indicators (pulse stretcher)

Asymmetrical RC Monostable (10 ms Hold Time) & Schmitt-Trigger Buffer



- Monostable hold time: τ ≈ 10 ms (1MΩ / 10nF + fast discharge diode)
- High-impedance tap: Does not load UART RX/TX lines

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Sheet: /InterfaceSheet/
File: Interface.kicad_sch

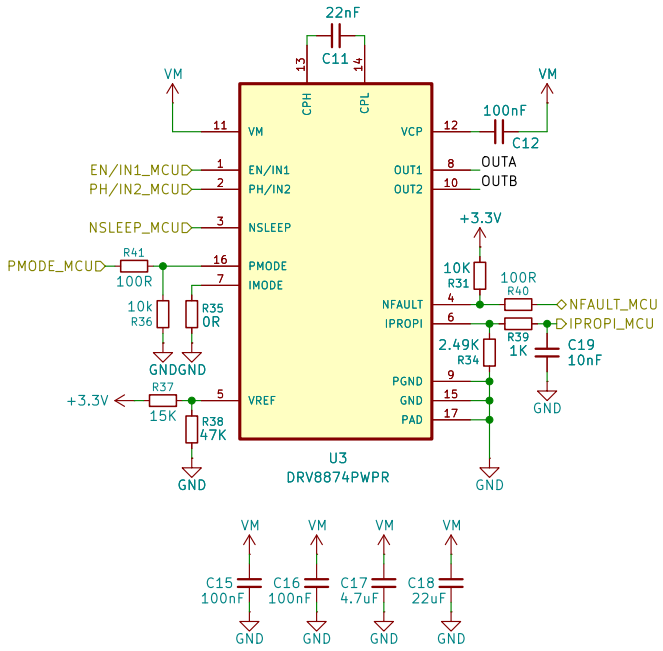
Title: KinetiCoreDC – Interface

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.2			Id: 2/5

Text

H-Bridge Motor Driver

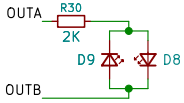
Integrated DC motor driver with internal current sensing and regulation



- Operating range: VM = 5.1V to 18V | I_peak = 6A
- Current limit (I_TRIP): ~2.6A (VREF = 2.50V via 15k/47k divider, R_IPROPI = 2.49kΩ)
- Current sense (IPROPI): 455 µA/A → ~1.13 V/A (RC filter fc ≈ 16 kHz)
- Control mode: PMODE selectable | NFAULT: Open-drain (10kΩ pull-up)

Output Status Indicator

Bi-directional LED polarity and activity indicator across H-bridge outputs



- Antiparallel LEDs across OUTA/OUTB (2kΩ ballast)
- Indicates rotation direction (polarity) and PWM duty cycle (brightness)

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Sheet: /DriverMotorSheet/
File: DriverMotor.kicad_sch

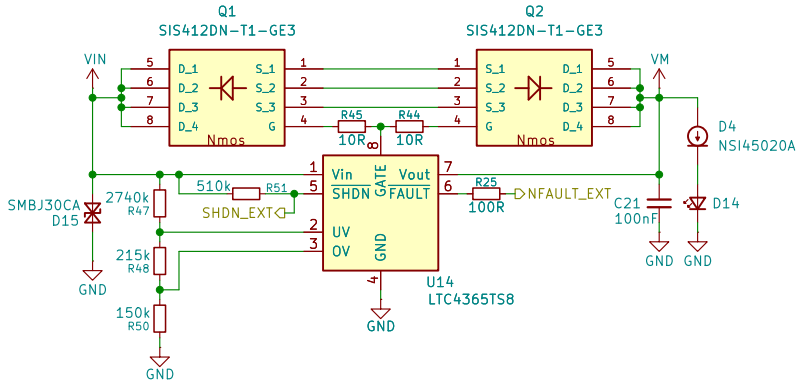
Title: KinetiCoreDC – Driver Motor

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.2			Id: 3/5

Text

External Supply Protection

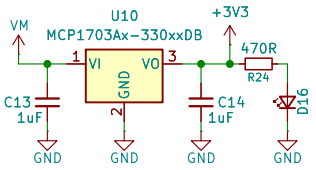
Overvoltage, Undervoltage, Reverse, and ESD Input Protection



- Window thresholds: UVLO = 5.08V | OVLO = 18.0V (Recovery delay: 36 ms)
- Protection: Reverse polarity to −40V (Back-to-back N-MOS) & 30V TVS clamp
- SHDN_EXT: Active-LOW (<0.4V = Sleep <10µA, 510kΩ pull-up to VIN)
- NFAULT_EXT: Open-drain active-LOW (Requires external pull-up on host board)

Supply LDO Regulator

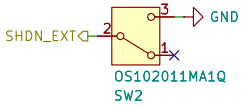
Low-quiescent-current 3.3V linear regulator for digital control logic



- Output: +3.3V ±0.4% | I_max = 250 mA | I_q = 2 µA
- Input: Supplied from protected VM rail (Max 16V continuous)

Manual Power Enable

On-board SPDT slide switch for manual power shutdown and sleep mode (< 10 µA).



- ON (Pin 1 – NC): Pulled HIGH to VIN via 510 kΩ (Normal operation)
- OFF (Pin 3 – GND): Pulls SHDN_EXT LOW (< 0.4 V), disabling input MOSFETs (< 10 µA sleep)

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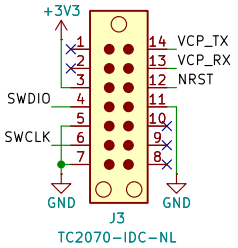
Sheet: /PowerSheet/
File: Power.kicad_sch

Title: KinetiCoreDC – Power

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.2			Id: 4/5

SWD Programming & UART Debug Interface

SWD flashing and VCP serial console connector
compatible with STDC14 / STLINK-V3



- Footprint: Tag-Connect TC2070-IDC-NL (No-Legs, spring-pin probe)
- Interfaces: SWD (SWDIO/SWCLK) + VCP UART (USART2) + Target NRST

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Sheet: /ConnectorSheet/
File: Connector.kicad_sch

Title: KinetiCoreDC – Connector

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.2			Id: 5/5

Text